

# Shallow dive into Linux kernel fuzzing with syzkaller

장선우

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# What is Fuzzing

**Fuzzing** : **Automated** software testing technique that puts **random inputs** into a computer program **to find bugs**.

- 자동화 : 무작위 input → 실행 → crash 체크
- Every crash from fuzzing ≠ security vulnerabilities
- Low cost, but high CPU consumption

# What is syzkaller

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**syzkaller** : An unsupervised **coverage-guided kernel fuzzer** initially developed by Google in 2015.

- 현재까지 Fixed Bugs : 4500+
- input 생성 → 실행 → coverage 측정 → crash 체크 → input 갱신
- Kernel fuzzing의 input = system call (entry point to kernel)
- Linux kernel feature 활용: KASAN(메모리 오류 감지), KCOV(코드 커버리지 측정) 등

# Key Components of syzkaller

## Core Binaries:

- syz-manager : Main 프로그램
- syz-executor : VM 내부에서 syscall 프로그램 실행

## Analysis Tools:

- syz-prog2c : syzkaller 프로그램 → C 코드 변환
- syz-db : corpus(input 프로그램)와 crash DB 관리

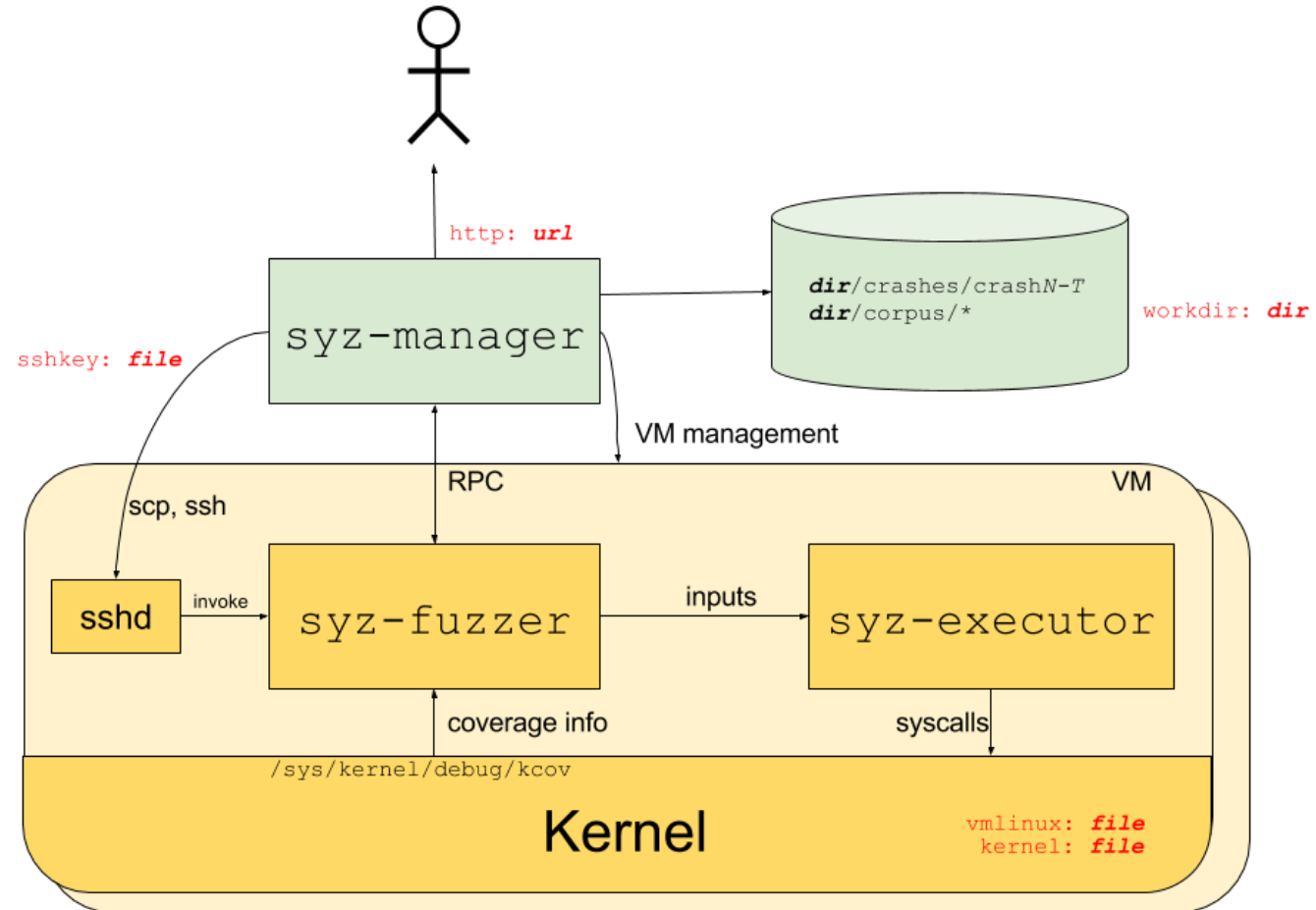


Image source: <https://lwn.net/Articles/677764/>

# What is syzbot

- syzkaller 대시보드:

Google이 운영하는 free open source service

- 24/7 자동 fuzzing + 버그 리포팅

- <https://syzkaller.appspot.com/upstream>

syzbot
Linux

Open [1496]
 Subsystems
 Fixed [6583]
 Invalid [16838]
 Missing Backports [204]
 Crashes
 Graphs
 Coverage
 Send us feedback

| Instances <a href="#">[tested repos]</a> : |             |        |        |                                   |         |         |                              |                         |           |        |
|--------------------------------------------|-------------|--------|--------|-----------------------------------|---------|---------|------------------------------|-------------------------|-----------|--------|
| Name                                       | Last active | Uptime | Corpus | Coverage <input type="checkbox"/> | Crashes | Execs   | Commit                       | Config                  | Freshness | Status |
| <a href="#">ci-qemu-gce-upstream-auto</a>  | now         | 21h33m | 26896  | <a href="#">433510</a>            | 257     | 840704  | <a href="#">46a51f4f5eda</a> | <a href="#">.config</a> | 1d07h     |        |
| <a href="#">ci-qemu-native-arm64-kvm</a>   | now         | 22h46m | 1477   | <a href="#">21330</a>             | 75      | 54509   | <a href="#">caf06487495f</a> | <a href="#">.config</a> | 1d05h     |        |
| <a href="#">ci-qemu-upstream</a>           | now         | 21h57m | 23463  | <a href="#">371824</a>            | 697     | 1907131 | <a href="#">46a51f4f5eda</a> | <a href="#">.config</a> | 1d07h     |        |
| <a href="#">ci-qemu-upstream-386</a>       | now         | 22h28m | 51436  | <a href="#">720703</a>            | 359     | 1450262 | <a href="#">46a51f4f5eda</a> | <a href="#">.config</a> | 1d07h     |        |

| open (1211):                                                                                     |                               |       |              |            |       |        |                        |             |          |
|--------------------------------------------------------------------------------------------------|-------------------------------|-------|--------------|------------|-------|--------|------------------------|-------------|----------|
| Title                                                                                            | Rank <input type="checkbox"/> | Repro | Cause bisect | Fix bisect | Count | Last   | Reported               | Discussions |          |
| <a href="#">INFO: task hung in acct</a> <a href="#">[kernel]</a>                                 | 1                             | C     |              |            | 1     | 3h55m  | <a href="#">2h41m</a>  |             |          |
| <a href="#">WARNING in ext4_xattr_block_set(3)</a> <a href="#">[ext4]</a>                        | -1                            | C     | error        |            | 12    | 7h32m  | <a href="#">2h41m</a>  |             |          |
| <a href="#">INFO: rcu detected stall in ext4_file_mmap_prepare</a> <a href="#">[ext4]</a>        | 1                             | C     |              |            | 1     | 12h27m | <a href="#">2h42m</a>  |             |          |
| <a href="#">general protection fault in find_match(6)</a> <a href="#">[net]</a>                  | 17                            | C     |              |            | 100   | 4d06h  | <a href="#">4h54m</a>  |             |          |
| <a href="#">KASAN: slab-use-after-free Read in napi_gro_frags(2)</a> <a href="#">[net]</a>       | 19                            | C     | done         |            | 3     | 7d18h  | <a href="#">5h04m</a>  |             |          |
| <a href="#">general protection fault in lec_arp_clear_vccs</a> <a href="#">[net]</a>             | 2                             |       |              |            | 1     | 4d12h  | <a href="#">10h11m</a> |             |          |
| <a href="#">possible deadlock in do_writepages(2)</a> <a href="#">[ext4]</a>                     | 4                             | syz   |              |            | 7     | 23h48m | <a href="#">16h48m</a> |             |          |
| <a href="#">kernel BUG in free_one_page(2)</a> <a href="#">[mm]</a>                              | -1                            | C     |              |            | 4     | 4d15h  | <a href="#">17h10m</a> |             |          |
| <a href="#">KASAN: slab-use-after-free Read in bch2_copygc(2)</a> <a href="#">[bcachefs]</a>     | 19                            |       |              |            | 7     | 11h13m | <a href="#">23h04m</a> |             |          |
| <a href="#">WARNING in add_timer</a> <a href="#">[net]</a>                                       | -1                            |       |              |            | 1     | 5d02h  | <a href="#">1d02h</a>  |             |          |
| <a href="#">WARNING in do_check(2)</a> <a href="#">[bpf]</a>                                     | -1                            | C     | done         |            | 31    | 1d06h  | <a href="#">1d04h</a>  |             |          |
| <a href="#">WARNING in maybe_exit_scc</a> <a href="#">[bpf]</a>                                  | -1                            | C     | done         |            | 36    | 1d05h  | <a href="#">1d04h</a>  | PATCH       | [now]    |
| <a href="#">general protection fault in gfs2_print_dbg(3)</a> <a href="#">[gfs2]</a>             | 2                             | C     | error        |            | 2     | 2d19h  | <a href="#">1d04h</a>  | PATCH       | [1h39m]  |
| <a href="#">KASAN: stack-out-of-bounds Read in cp2112_xfer</a> <a href="#">[input]</a>           | 17                            | C     |              |            | 4     | 2d15h  | <a href="#">1d04h</a>  | PATCH       | [17h40m] |
| <a href="#">INFO: task hung in erofs_read</a> <a href="#">[erofs]</a>                            | 1                             | C     |              |            | 4     | 1d10h  | <a href="#">1d04h</a>  | PATCH       | [14h02m] |
| <a href="#">KASAN: slab-use-after-free Write in iommufd_eventq_fops_release ...</a>              | 24                            | C     |              |            | 5     | 3d11h  | <a href="#">1d04h</a>  |             |          |
| <a href="#">INFO: trying to register non-static key in ntfs_set_size</a> <a href="#">[ntfs3]</a> | -1                            | C     | done         |            | 6     | 2d04h  | <a href="#">1d04h</a>  | PATCH       | [17h00m] |

# Input & Output of syzkaller

```
sunwoo@fedora:~/sources/syzkaller/build (master *)$ cat trixie.cfg
{
  "target": "linux/amd64",
#  "http": "127.0.0.1:56741",
  "http": "0.0.0.0:56741",
  "workdir": "/home/sunwoo/sources/syzkaller/build/workdir_250916",
  "kernel_obj": "/home/sunwoo/sources/linux-6.16",
  "image": "/home/sunwoo/sources/syzkaller/build/trixie.img",
  "sshkey": "/home/sunwoo/sources/syzkaller/build/trixie.id_rsa",
  "syzkaller": "/home/sunwoo/sources/syzkaller",
  "procs": 8,
  "type": "qemu",
  "vm": {
    "count": 4,
    "kernel": "/home/sunwoo/sources/syzkaller/build/arch/x86/boot/bzImage",
    "cpu": 2,
    "mem": 2048,
    "cmdline": "console=ttyS0 root=/dev/sda net.ifnames=0"
  }
}
```

```
sunwoo@fedora:~/sources/syzkaller/build (master *)$ tree workdir_250916/
workdir_250916/
├── corpus.db
├── crashes
│   ├── 993b6f7591540f332cfbe334e31ad0bbc13fbde7
│   │   ├── description
│   │   ├── log0
│   │   ├── machineInfo0
│   │   ├── report0
│   │   └── title-stat
│   └── e6109b119e6879abe37e0a2782d8deb38764a11b
│       ├── description
│       ├── log0
│       ├── log1
│       ├── machineInfo0
│       ├── machineInfo1
│       ├── report0
│       ├── report1
│       └── title-stat
└── instance-lock
```

\$ syz-manager --debug --config=./trixie.cfg

# Hands-on:

syzkaller



stats

coverage

syscalls

corpus

VMs

config

|             |                        |
|-------------|------------------------|
| candidates  | 0                      |
| corpus      | <a href="#">14432</a>  |
| coverage    | <a href="#">106446</a> |
| exec total  | 23073397 (57/sec)      |
| pending     | 0                      |
| reproducing | 0                      |
| crash types | 7                      |
| crashes     | 8535                   |
| suppressed  | 0                      |
| syscalls    | <a href="#">2271</a>   |
| uptime      | 401498 sec             |

## Crashes:

| Description                                                                    | Rank | Count | First Time       | Last Time        | Report                           | Repro Attempt Count |
|--------------------------------------------------------------------------------|------|-------|------------------|------------------|----------------------------------|---------------------|
| <a href="#">failed to read from qemu: EOF</a>                                  | -1   | 4     | 2025/09/08 11:47 | 2025/09/08 11:47 |                                  |                     |
| <a href="#">KASAN: out-of-bounds Read in ext4_xattr_set_entry</a>              | 17   | 1     | 2025/09/12 07:04 | 2025/09/12 07:04 |                                  | 1                   |
| <a href="#">lost connection to test machine</a>                                | -1   | 3     | 2025/09/08 11:50 | 2025/09/08 11:50 |                                  |                     |
| <a href="#">no output from test machine</a>                                    | -1   | 100   | 2025/09/13 10:39 | 2025/09/13 10:39 | <a href="#">non-reproducible</a> | 3                   |
| <a href="#">SYZFAIL: ehtable checkpoint: socket(AF_INET, SOCK_STREAM, I...</a> | -1   | 1     | 2025/09/12 14:00 | 2025/09/12 14:00 |                                  | 1                   |
| <a href="#">SYZFAIL: mmap of data segment failed</a>                           | -1   | 100   | 2025/09/13 10:39 | 2025/09/13 10:39 | <a href="#">non-reproducible</a> | 3                   |
| <a href="#">WARNING in corrupted</a>                                           | -1   | 66    | 2025/09/13 08:56 | 2025/09/13 08:56 |                                  |                     |
| <a href="#">WARNING in rcu_torture_writer</a>                                  | -1   | 100   | 2025/09/13 10:38 | 2025/09/13 10:38 | <a href="#">has C repro</a>      | 3                   |
| <a href="#">WARNING in rcutorture_oom_notify</a>                               | -1   | 100   | 2025/09/13 09:39 | 2025/09/13 09:39 | <a href="#">has C repro</a>      | 3                   |

## Log:

```
2025/09/13 10:39:24 VM 0: crash(tail10): SYZFAIL: mmap of data segment failed
2025/09/13 10:39:24 VM 0: crash(tail11): SYZFAIL: mmap of data segment failed
2025/09/13 10:39:24 VM 0: crash(tail12): SYZFAIL: mmap of data segment failed
2025/09/13 10:39:24 VM 0: crash(tail13): SYZFAIL: mmap of data segment failed
2025/09/13 10:39:24 VM 0: crash(tail14): SYZFAIL: mmap of data segment failed
2025/09/13 10:39:24 VM 0: crash(tail15): SYZFAIL: mmap of data segment failed
2025/09/13 10:39:24 VM 0: crash(tail16): SYZFAIL: mmap of data segment failed
2025/09/13 10:39:24 VM 0: crash(tail17): SYZFAIL: mmap of data segment failed
2025/09/13 10:39:24 VM 0: crash(tail18): SYZFAIL: mmap of data segment failed
2025/09/13 10:39:24 VM 0: crash(tail19): SYZFAIL: mmap of data segment failed
2025/09/13 10:39:24 VM 0: crash(tail20): SYZFAIL: repeatedly failed to execute the program
2025/09/13 10:39:28 candidates=0 corpus=14430 coverage=106444 exec total=23071730 (57/sec) pending=0 reproducing=0
2025/09/13 10:39:38 candidates=0 corpus=14430 coverage=106444 exec total=23071998 (57/sec) pending=0 reproducing=0
2025/09/13 10:39:41 runner 0 connected
2025/09/13 10:39:48 candidates=0 corpus=14430 coverage=106444 exec total=23072310 (57/sec) pending=0 reproducing=0
2025/09/13 10:39:49 VM 3: crash: no output from test machine
2025/09/13 10:39:58 candidates=0 corpus=14431 coverage=106444 exec total=23072978 (57/sec) pending=0 reproducing=0
```

syzkaller



stats

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KASAN: out-of-bounds Read in ext4\_xattr\_set\_entry

| # | Log                 | Report                 | Time             | Tag |
|---|---------------------|------------------------|------------------|-----|
| 0 | <a href="#">log</a> | <a href="#">report</a> | 2025/09/12 07:04 |     |

# Hands-on: Found a bug from my syzkaller in a week

*KASAN: out-of-bounds Read in ext4\_xattr\_set\_entry*

```
1 Call Trace:
2  <TASK>
3  __dump_stack lib/dump_stack.c:94 [inline]
4  dump_stack_lvl+0x71/0xa0 lib/dump_stack.c:120
5  print_address_description mm/kasan/report.c:378 [inline]
6  print_report+0x174/0x4f6 mm/kasan/report.c:482
7  kasan_report+0xce/0x100 mm/kasan/report.c:595
8  check_region_inline mm/kasan/generic.c:175 [inline]
9  kasan_check_range+0x10f/0x1e0 mm/kasan/generic.c:189
10 __asan_memmove+0x23/0x60 mm/kasan/shadow.c:94
11 ext4_xattr_set_entry+0x10c5/0x1a20 fs/ext4/xattr.c:1751
12 ext4_xattr_ibody_set+0x377/0x570 fs/ext4/xattr.c:2263
13 ext4_destroy_inline_data_nolock+0x22c/0x520 fs/ext4/inline.c:455
14 ext4_convert_inline_data_nolock+0x15a/0xd30 fs/ext4/inline.c:1140
15 ext4_convert_inline_data+0x498/0x570 fs/ext4/inline.c:1997
16 ext4_fallocate+0xe4/0x2a40 fs/ext4/extents.c:4756
17 vfs_fallocate+0x32a/0xba0 fs/open.c:341
18 ksys_fallocate fs/open.c:365 [inline]
19 __do_sys_fallocate fs/open.c:370 [inline]
20 __se_sys_fallocate fs/open.c:368 [inline]
21 __x64_sys_fallocate+0xc4/0x130 fs/open.c:368
22 do_syscall_x64 arch/x86/entry/syscall_64.c:63 [inline]
23 do_syscall_64+0xa4/0x260 arch/x86/entry/syscall_64.c:94
24 entry_SYSCALL_64_after_hwframe+0x77/0x7f
```

- **트리거**: `fallocate(fd, 0, 7, 0x240faa3)` 시스템 콜 호출 시, 파일의 inline data를 변환하는 과정.
- **원인**: `ext4_xattr_set_entry` 함수에서 Integer Underflow. 이로 인해 `memmove` 함수의 크기가 -16으로 계산.
- **재현 x**: 특정한 조건의 corrupted ext4 filesystem 필요(?)



**감사합니다!**

# References

## [Install and Build]

<https://github.com/google/syzkaller/blob/master/docs/linux/setup.md>

[https://github.com/google/syzkaller/blob/master/docs/linux/setup\\_ubuntu-host\\_qemu-vm\\_x86-64-kernel.md](https://github.com/google/syzkaller/blob/master/docs/linux/setup_ubuntu-host_qemu-vm_x86-64-kernel.md)

<https://github.com/google/syzkaller/blob/master/tools/create-image.sh>

[https://github.com/google/syzkaller/blob/master/docs/linux/kernel\\_configs.md](https://github.com/google/syzkaller/blob/master/docs/linux/kernel_configs.md)

<https://github.com/google/syzkaller/blob/master/pkg/mgrconfig/testdata/qemu-example.cfg>

## [How to]

<https://github.com/google/syzkaller/blob/master/docs/usage.md>

[https://github.com/google/syzkaller/blob/master/docs/reproducing\\_crashes.md](https://github.com/google/syzkaller/blob/master/docs/reproducing_crashes.md)

## [Etc]

<https://lwn.net/Articles/677764/>

<https://lwn.net/Articles/749910/>

<https://github.com/google/syzkaller/blob/master/docs/syzbot.md>

Tech talks: <https://github.com/google/syzkaller/blob/master/docs/talks.md>

Research: <https://github.com/google/syzkaller/blob/master/docs/research.md>